PEPstrMOD2 Manual

1. Introduction

PEPstrMOD2 is an advanced computational tool designed for the prediction of tertiary structures of peptides containing chemical, structural, and post-translational modifications. It represents a significant enhancement over the original PEPstrMOD framework, addressing previous limitations in scalability and flexibility.

The updated platform supports both standalone execution and Docker-based deployment, enabling reproducible and efficient large-scale analyses. Additionally, PEPstrMOD2 incorporates modern large language model (LLM)-based approaches to improve structural prediction accuracy, particularly for peptides with complex or multiple modifications.

This manual provides detailed instructions for input preparation, command-line usage, and interpretation of outputs.

For queries or feedback, please contact: raghava@iiitd.ac.in

2. Modified Amino Acid/Peptide (MAP) Format

2.1 Overview

PEPstrMOD2 utilizes the Modified Amino Acid/Peptide (MAP) format, an extension of the standard FASTA format. This format enables explicit annotation of residue-level modifications, ensuring accurate structural modeling.

2.2 Format Structure

- Each entry begins with a header line (> identifier)
- The peptide sequence uses standard one-letter amino acid codes
- Modifications are specified immediately after residues using curly braces {}

2.3 Syntax

Residue{type:descriptor}

Type	**Description**	**Example**
nnr	Non-natural residues	DAEC{nnr:2ZC}RM
ptm	Post-translational modifications	ACS{ptm:phos}DERF
nterm	N-terminal modifications	AACDEFGH{nt:Acet}
cterm	C-terminal modifications	GACDEFGY{ct:Amid}
cyc	Cyclization	ACDECCHIKLC{cyc:2-5} ACDECCHIKLC{cyc:N-C}
d	D-amino acid	GACDE{d}GH

2.4 Combined Modifications

Multiple modifications can be applied within a single sequence.
Example: K{ptm:Coxy}CD{d}CF{nnr:X5U}GHIK{nt:Acet}{cyc:2-4}

2.5 Backward Compatibility

Removing all modification annotations ({}) converts the sequence into standard FASTA format, ensuring compatibility with other tools. In case of nnr, corresponding parent residue is inserted.

3. Supported Input Types

PEPstrMOD2 accepts the following input formats:

1. Single MAP sequence (quoted string)
   Example: "GA{d}CDEFGH"

2. Text or FASTA file containing multiple sequences
   Example:
   >Seq1
   GA{d}CDEFGH

>Seq2
ACD{ptm:Beta}FGHIK

4. Command-Line Usage

4.1 Basic Syntax

python pepstrmod2.py [OPTIONS]

Option	Description
-h, --help	Show help message and exit
-i INPUT, --input INPUT	Single MAP sequence in quotes OR a text/fasta file with multiple sequences.
-p PREFIX, --prefix PREFIX	Prefix for output filenames. Default: test.
-o OUTDIR, --outdir OUTDIR	Main output folder. Default: output
-e {vac,phil,phob}, --env {vac,phil,phob}	Simulation environment: vacuum or hydrophobic/hydrophilic. Default: phil.
-s SIM, --sim SIM	Simulation time in picoseconds. Default: 100 ps
-t {yes,no}, --traj {yes,no}	Perform trajectory analysis. Default: no
-c {yes,no}, --cluster {yes,no}	Perform cluster analysis. Default: no
-g {yes,no}, --graph {yes,no}	Generate energy plots and RMSD graphs. Default: no
-m {esmfold,colabfold}, --method {esmfold,colabfold}	Structure prediction method. Default: alphafold2

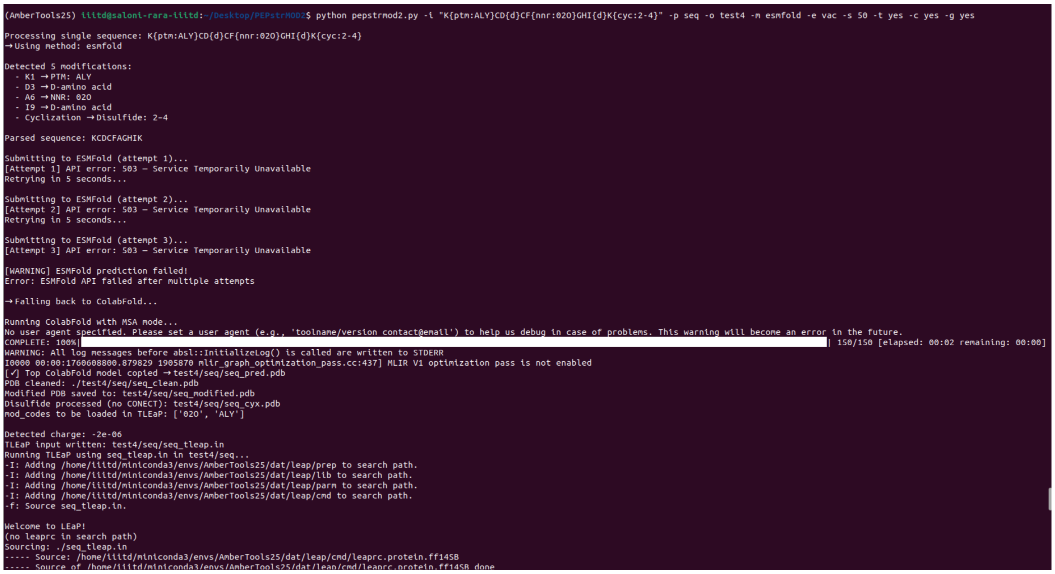
(AmberTools25) iiitd@saloni-rara-iiitd:~/Desktop/PEPstrMOD2$ **python pepstrmod2.py --h**

usage: pepstrmod2.py [-h] -i INPUT [-p PREFIX] [-o OUTDIR] [-e {vac,phil}] [-s SIM] [-t {yes,no}] [-c {yes,no}] [-g {yes,no}] [-m {esmfold,colabfold}]

5. Example

Command:

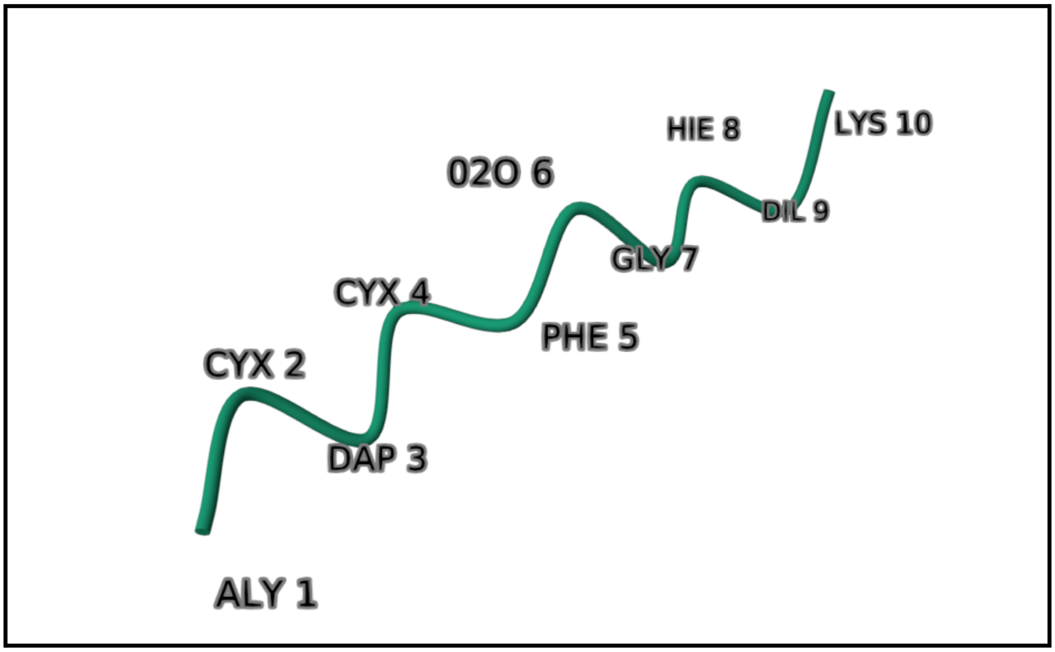
python pepstrmod2.py -i "K{ptm:Phos}CD{d}CF{nnr:02O}GHI{d}K{cyc:2-4}" -p seq -o test4 -m esmfold -e vac -s 50 -t yes -c yes -g yes



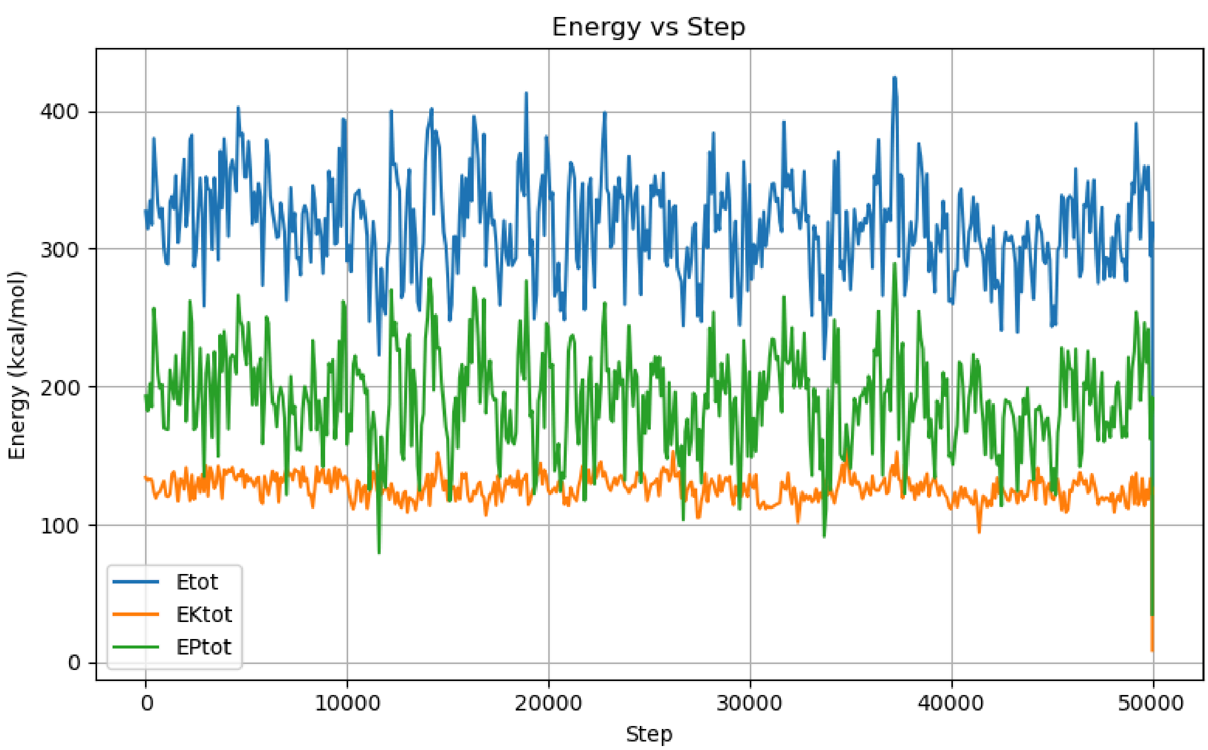
```
(AmberTools25) iiitd@saloni-rara-iiitd:~/Desktop/PEPstrMOD2$ python pepstrmod2.py -i "K{ptm:ALY}CD{d}CF{nnr:02O}GHI{d}K{cyc:2-4}" -p seq -o test4 -m esmfold -e vac -s 50 -t yes -c yes -g yes

Processing single sequence: K{ptm:ALY}CD{d}CF{nnr:02O}GHI{d}K{cyc:2-4}
→Using method: esmfold

Detected 5 modifications:
  - K1 →PTM: ALY
  - D3 →D-amino acid
  - A6 →NNR: 02O
  - I9 →D-amino acid
  - Cyclization →Disulfide: 2-4

Parsed sequence: KCDCFAGHIK

Submitting to ESMFold (attempt 1)...
[Attempt 1] API error: 503 – Service Temporarily Unavailable
Retrying in 5 seconds...

Submitting to ESMFold (attempt 2)...
[Attempt 2] API error: 503 – Service Temporarily Unavailable
Retrying in 5 seconds...

Submitting to ESMFold (attempt 3)...
[Attempt 3] API error: 503 – Service Temporarily Unavailable

[WARNING] ESMFold prediction failed!
Error: ESMFold API failed after multiple attempts

→Falling back to ColabFold...

Running ColabFold with MSA mode...
No user agent specified. Please set a user agent (e.g., 'toolname/version contact@email') to help us debug in case of problems. This warning will become an error in the future.
COMPLETE: 100%|                                                            | 150/150 [elapsed: 00:02 remaining: 00:00]
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1760608800.879829 1905870 mlir_graph_optimization_pass.cc:437] MLIR V1 optimization pass is not enabled
[✓] Top ColabFold model copied →test4/seq/seq_pred.pdb
PDB cleaned: ./test4/seq/seq_clean.pdb
Modified PDB saved to: test4/seq/seq_modified.pdb
Disulfide processed (no CONECT): test4/seq/seq_cyx.pdb
mod_codes to be loaded in TLEaP: ['02O', 'ALY']

Detected charge: -2e-06
TLEaP input written: test4/seq/seq_tleap.in
Running TLEaP using seq_tleap.in in test4/seq...
-I: Adding /home/iiitd/miniconda3/envs/AmberTools25/dat/leap/prep to search path.
-I: Adding /home/iiitd/miniconda3/envs/AmberTools25/dat/leap/lib to search path.
-I: Adding /home/iiitd/miniconda3/envs/AmberTools25/dat/leap/parm to search path.
-I: Adding /home/iiitd/miniconda3/envs/AmberTools25/dat/leap/cmd to search path.
-f: Source seq_tleap.in.

Welcome to LEaP!
(no leaprc in search path)
Sourcing: ./seq_tleap.in
----- Source: /home/iiitd/miniconda3/envs/AmberTools25/dat/leap/cmd/leaprc.protein.ff14SB
----- Source of /home/iiitd/miniconda3/envs/AmberTools25/dat/leap/cmd/leaprc.protein.ff14SB done
```

The final output will be saved as {prefix}_final.pdb.

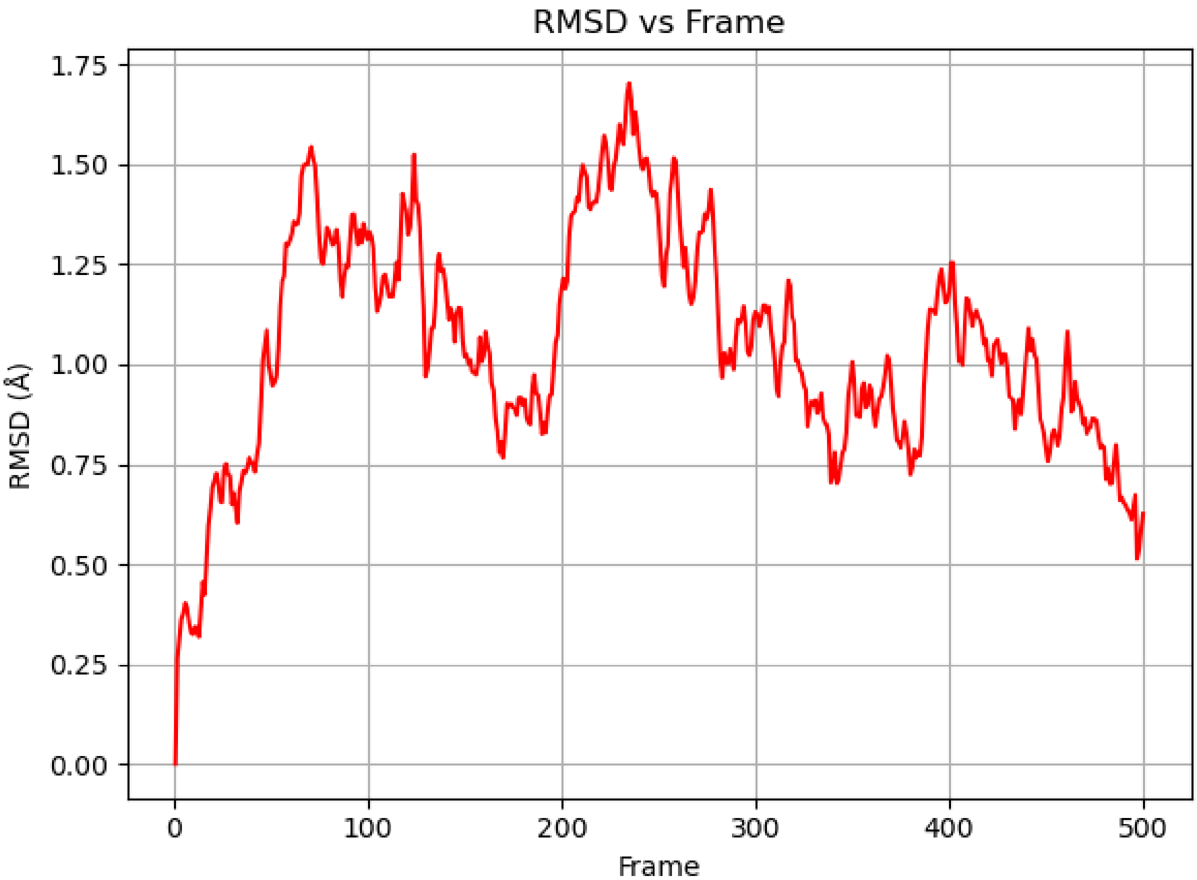
The final pdb file is visualized in the RCSB PDB viewer.



Energy Graph



RMSD Graph


RMSD vs Frame
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
RMSD (Å)
0
100
200
300
400
500
Frame